

BS-CH101 / BS-CH201 — Chemistry-I

Semester I • 3L:1T:0P • 4 Credits

Module 1 Exam Notes: Atomic and Molecular Structure

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Atomic and Molecular Structure

Atomic and molecular structure explains the arrangement of electrons in atoms and molecules, the origin of bonding, and the connection between energy levels and observable properties.

1.1 Schrodinger Wave Equation

In quantum chemistry, an electron is described by a wave function ψ . The square of the wave function gives probability density, so $|\psi|^2 d\tau$ is the probability of finding the electron in a small volume element $d\tau$.

Time-Independent Schrodinger Equation

$$\hat{H}\psi = E\psi$$

where $\hat{H}$ is the Hamiltonian operator, ψ is the wave function, and E is the permitted energy. For a one-dimensional particle,

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi}{dx^2} + V\psi = E\psi$$

1.2 Particle in a One-Dimensional Box

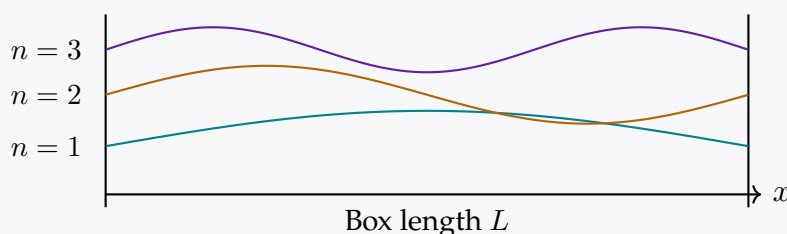
For a particle confined between $x = 0$ and $x = L$, the potential energy is zero inside the box and infinite outside it. Boundary conditions are:

$$\psi(0) = 0, \quad \psi(L) = 0$$

The allowed wave functions and energies are:

$$\psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, \quad E_n = \frac{n^2 h^2}{8mL^2}, \quad n = 1, 2, 3, \dots$$

Particle in a Box: Quantized Standing Waves



1.3 Molecular Orbitals of Diatomic Molecules

Molecular orbital (MO) theory states that atomic orbitals combine to form molecular orbitals spread over the whole molecule. Constructive overlap gives bonding MOs and destructive overlap gives antibonding MOs.

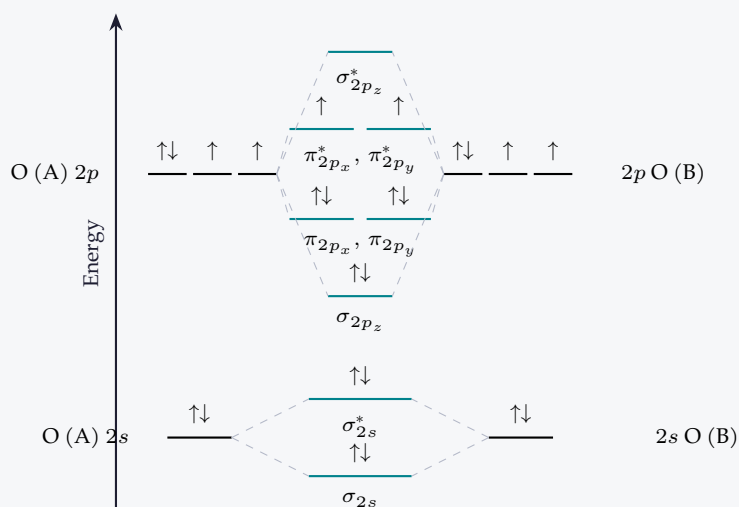
Bond Order

$$\text{Bond order} = \frac{N_b - N_a}{2}$$

where N_b is the number of electrons in bonding MOs and N_a is the number of electrons in antibonding MOs. Higher bond order means shorter, stronger bond.

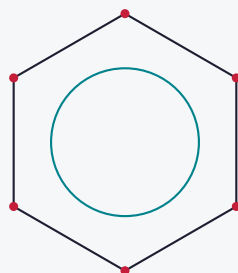
Important examples:

Molecule	Configuration idea	Bond order	Magnetism
H ₂	Two electrons in σ_{1s}	1	Diamagnetic

Molecular Orbital Energy Level Diagram of Oxygen (O₂)**1.4 Pi-Molecular Orbitals and Aromaticity**

Pi-electrons in conjugated systems are delocalized. Benzene has six pi-electrons distributed over six carbon atoms, making it unusually stable.

Aromatic compound: A cyclic, planar, fully conjugated molecule having $(4n + 2)$ pi-electrons, where $n = 0, 1, 2, \dots$

Benzene Pi-Electron Delocalization

6 delocalized pi-electrons satisfy Huckel rule

1.5 Pi-Molecular Orbitals of 1,3-Butadiene

1,3-butadiene ($\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$) has a conjugated system of four p orbitals containing four π electrons. These combine to form four π molecular orbitals ($\psi_1, \psi_2, \psi_3, \psi_4$) with increasing energy and nodes.

- **Nodal Properties:** The number of nodes in orbital ψ_k is equal to $(k - 1)$. A node is a position where the wave function changes sign and electron probability is zero.
- **HOMO/LUMO:** In the ground state, the four π electrons fill the two lowest-energy orbitals (ψ_1, ψ_2). Thus, ψ_2 is the **Highest Occupied Molecular Orbital (HOMO)** and ψ_3 is the **Lowest Unoccupied Molecular Orbital (LUMO)**.

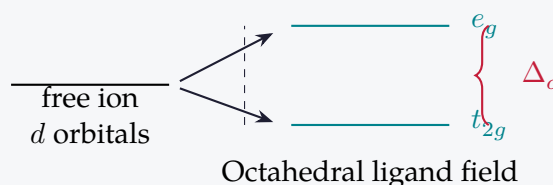
Pi-Molecular Orbitals of 1,3-Butadiene



1.6 Crystal Field Theory

In transition-metal complexes, surrounding ligands split the five degenerate d orbitals into different energy groups. The splitting controls colour, magnetic behaviour, and stability.

Octahedral Crystal Field Splitting



1.7 Band Structure and Doping

In solids, many atomic orbitals combine to form bands. Conductivity depends on the band gap E_g between valence band and conduction band.

Material	Band gap	Electrical behaviour
Conductor	Overlapping bands or very small gap	Electrons move easily
Semiconductor	Moderate gap, about 1–3 eV	Conductivity increases by heat or doping
Insulator	Large gap, usually > 5 eV	Very poor conductor

Exam Points

- Derive $E_n = n^2 h^2 / (8mL^2)$ using boundary conditions.
- For MO questions, always calculate bond order and then state magnetic nature.
- For CFT, mention splitting pattern, Δ_o , high spin or low spin, and magnetic moment.
- For band theory, draw valence band, conduction band, and band gap.

2. Exam-Oriented Derivations and Applications**2.1 Derivation of Particle in a Box Energy**

For a one-dimensional box, potential energy inside the box is zero. Therefore the Schrodinger equation becomes:

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi}{dx^2} = E\psi$$

Rearranging,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2mE}{h^2}\psi = 0$$

Let

$$k^2 = \frac{8\pi^2mE}{h^2}$$

Then the differential equation becomes:

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0$$

Its general solution is:

$$\psi = A \sin kx + B \cos kx$$

Boundary condition at $x = 0$ gives $B = 0$. Boundary condition at $x = L$ gives:

$$A \sin kL = 0$$

For a non-zero wave function:

$$kL = n\pi, \quad k = \frac{n\pi}{L}$$

Substituting in $k^2 = 8\pi^2mE/h^2$:

$$E_n = \frac{n^2 h^2}{8mL^2}$$

Solved Example: Energy Level Ratio

Problem: Find the ratio of the first three energy levels for a particle in a box.

Solution: Since $E_n \propto n^2$,

$$E_1 : E_2 : E_3 = 1^2 : 2^2 : 3^2 = 1 : 4 : 9$$

Exam note: Energy gaps are not equal; they increase with n .

2.2 MO Filling Order and Magnetic Behaviour

For second-period diatomic molecules, MO ordering changes after N_2 because s - p mixing becomes less important. This is a common exam trap.

Molecules	MO order to remember	Important result
B_2, C_2, N_2	$\sigma_{2s} < \sigma_{2s}^* < \pi_{2p} < \sigma_{2p} < \pi_{2p}^* < \sigma_{2p}^*$	B_2 is paramagnetic; N_2 has bond order 3
O_2, F_2, Ne_2	$\sigma_{2s} < \sigma_{2s}^* < \sigma_{2p} < \pi_{2p} < \pi_{2p}^* < \sigma_{2p}^*$	O_2 is paramagnetic due to two unpaired electrons

Solved Example: Bond Order of Oxygen

For O_2 , total valence electrons are 12.

$$N_b = 8, \quad N_a = 4$$

$$\text{Bond order} = \frac{8 - 4}{2} = 2$$

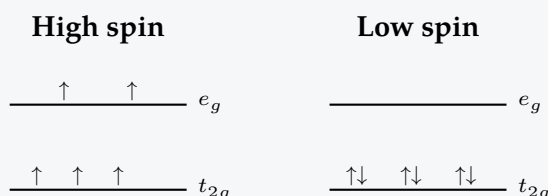
Two electrons remain unpaired in π_{2p}^* orbitals, so oxygen is **paramagnetic**.

2.3 Crystal Field Splitting: High Spin and Low Spin

In octahedral complexes, electrons first occupy lower t_{2g} orbitals. Whether pairing occurs depends on the relative values of crystal field splitting energy Δ_o and pairing energy P .

Case	Condition	Result
High spin complex	$\Delta_o < P$	Electrons avoid pairing; more unpaired electrons; usually weak-field ligands
Low spin complex	$\Delta_o > P$	Electrons pair in t_{2g} before occupying e_g ; fewer unpaired electrons; usually strong-field ligands

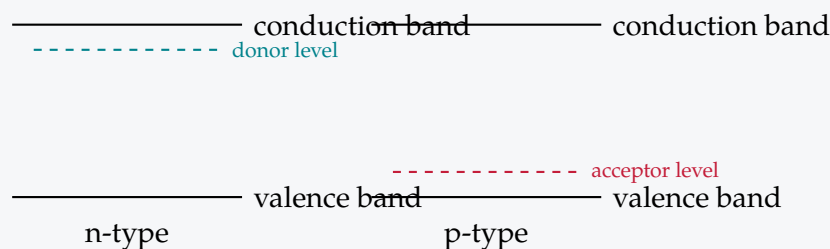
High Spin and Low Spin Choice for Octahedral Complexes



2.4 n-Type and p-Type Semiconductors

Doping adds controlled impurities to a semiconductor. Pentavalent dopants such as P or As introduce donor levels and produce n-type semiconductors. Trivalent dopants such as B or Al introduce acceptor levels and produce p-type semiconductors.

Doping Levels in Semiconductors



How to Write 10-Mark Answers in This Module

1. Start with a definition in 2–3 lines.
2. Draw the required energy-level or orbital diagram.
3. Write the governing formula with symbol meanings.
4. Add one example such as O_2 paramagnetism, benzene aromaticity, or n-type doping.
5. End with one application or exam conclusion.

2.5 Important Long-Answer: Aromaticity of Benzene

Benzene is the standard example of aromatic stability. A complete answer should not only state Huckel rule, but also connect structure, pi-electron count, planarity, and equal bond lengths.

1. Benzene is cyclic and planar, so all six carbon atoms can overlap through unhybridized p orbitals.
2. Each carbon contributes one pi-electron, giving a total of six pi-electrons.
3. Huckel rule requires $(4n + 2)$ pi-electrons. For benzene:

$$4n + 2 = 6 \Rightarrow n = 1$$

4. The pi-electrons are delocalized above and below the ring.
5. Delocalization makes all C–C bonds equivalent and gives extra stability.

Exam Comparison: Aromatic, Anti-Aromatic and Non-Aromatic

Type	Condition	Pi-electrons	Example idea
Aromatic	Cyclic, planar, conjugated	$(4n + 2)$	Benzene, cyclopentadienyl anion
Anti-aromatic	Cyclic, planar, conjugated	$4n$	Cyclobutadiene type systems
Non-aromatic	Fails planarity or conjugation	Rule not applicable	Non-planar or saturated ring

2.6 Important Long-Answer: Crystal Field Theory and Magnetism

Crystal field theory explains colour and magnetic properties of transition metal complexes by ligand-induced splitting of d orbitals.

Crystal Field Stabilization Energy

For an octahedral complex:

$$\text{CFSE} = (-0.4x + 0.6y)\Delta_o + pP$$

where x is the number of electrons in t_{2g} orbitals, y is the number of electrons in e_g orbitals, and pP is the pairing energy contribution.

Solved Example: Octahedral d^3 Configuration

For a d^3 octahedral ion, electrons occupy t_{2g} orbitals singly:



$$\text{CFSE} = 3(-0.4\Delta_o) = -1.2\Delta_o$$

There are three unpaired electrons, so the spin-only magnetic moment is:

$$\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87 \text{ BM}$$

2.7 Band Theory: Exam Numericals and Concepts

Band gap decides electrical behaviour. If the gap is small, thermal energy can excite electrons to the conduction band. Doping reduces the effective energy needed for charge conduction.

Question type	What to write	Common conclusion
Conductor vs semiconductor	Draw overlapping band and small band gap	Conductor has many free electrons
Intrinsic semiconductor	Pure semiconductor with thermally generated carriers	Electron and hole concentrations are equal
n-type semiconductor	Donor level near conduction band	Electrons are majority carriers
p-type semiconductor	Acceptor level near valence band	Holes are majority carriers

Previous-Year Style Questions

1. Derive the energy expression for a particle in a one-dimensional box and explain why $n = 0$ is not allowed.
2. Draw the MO diagram of O_2 and explain its paramagnetic nature.
3. Explain aromaticity of benzene using Huckel rule.
4. Explain high-spin and low-spin octahedral complexes with suitable examples.

5. Compare conductors, semiconductors and insulators using band theory.

Quick Revision — Module 1

Atomic and Molecular Structure | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$\hat{H}\psi = E\psi$	Schrodinger eigenvalue equation for allowed energies.
$E_n = \frac{n^2 h^2}{8mL^2}$	Energy of a particle in a one-dimensional box.
$\psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$	Normalized wave function for a particle in a box.
$\text{Bond order} = \frac{N_b - N_a}{2}$	Predicts bond strength and stability in MO theory.
$\mu = \sqrt{n(n+2)} \text{ BM}$	Spin-only magnetic moment for n unpaired electrons.
$4n + 2$	Huckel rule for aromatic pi-electron count.

One-Line Definitions

- **Wave function:** Mathematical function whose square gives probability density.
- **Node:** Point or surface where $\psi = 0$.
- **Bonding MO:** Lower-energy MO formed by constructive overlap.
- **Antibonding MO:** Higher-energy MO formed by destructive overlap.
- **Aromaticity:** Extra stability due to cyclic delocalization of $(4n + 2)$ pi-electrons.
- **Crystal field splitting:** Separation of metal d orbitals in a ligand field.
- **Band gap:** Energy gap between valence and conduction bands.

Frequently Asked / Viva Questions

Q1. State the Schrodinger equation.

The time-independent form is $\hat{H}\psi = E\psi$, where $\hat{H}$ is the energy operator.

Q2. Why is energy quantized for a particle in a box?

Only wave functions satisfying $\psi(0) = 0$ and $\psi(L) = 0$ are allowed, giving discrete n values.

Q3. What is the zero-point energy?

The minimum energy of a confined particle, $E_1 = h^2/(8mL^2)$.

Q4. Define bond order.

Bond order is $(N_b - N_a)/2$; it measures net bonding in a molecule.

Q5. Why is oxygen paramagnetic?

Frequently Asked / Viva Questions (contd.)

O₂ has two unpaired electrons in antibonding π^* orbitals.

Q6. What is Huckel rule?

A cyclic planar conjugated molecule is aromatic if it has $(4n + 2)$ pi-electrons.

Q7. Why is benzene stable?

Six pi-electrons are delocalized over the ring, satisfying Huckel rule with $n = 1$.

Q8. What is crystal field splitting?

Splitting of metal d orbitals into groups of different energy by ligands.

Q9. What is the role of doping in semiconductors?

Doping introduces donor or acceptor levels and increases carrier concentration.

Q10. How are conductors and insulators distinguished by band theory?

Conductors have overlapping or nearly touching bands, whereas insulators have a large band gap.

Summary Reference Table

Topic	Key idea	Exam output
Particle in box	Boundary conditions give discrete energies	Derivation plus energy-level diagram
MO theory	Atomic orbitals combine into bonding and antibonding MOs	Bond order and magnetism
CFT	Ligands split d orbitals	Splitting diagram and unpaired electrons
Band theory	Bands decide electrical conductivity	Conductor, semiconductor, insulator comparison